DD-T-710
OCTOBER 16, 1957

FEDERAL SPECIFICATION

TUBE, CONNECTING, LABORATORY GLASSWARE AND CONNECTOR, ELASTIC TUBING, BRANCHED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers right-angle and Y-shaped glass connecting tubes used for connecting laboratory and pharmaceutical glassware components, and rubber tubing.

1.2 Classification.—Glass connecting tubes shall be of the following types and sizes, as specified (see 6.1):

Type I.—Y-shaped.

Size A.—5 millimeter outside diameter.
Size B.—8 millimeter outside diameter.

Type II.—Right-angle.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP–B–566 — Boxes, Folding, Paperboard.
PPP–B–585—Boxes; Wood, Wirebound.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–B–10377—Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

441883/57/No. 124

FSC 6640

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MIL–STD–129 — Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

U. S. Pharmacopoeial Convention Publication:

Pharmacopoeia of the United States.

(Application for copies should be addressed to the Mack Printing Company, Easton, Pa.)

3. REQUIREMENTS

3.1 Material.—Material shall be borosilicate glass of a coefficient of expansion not exceeding 0.0000055 cm/cm/1° C. from 0° to 310° C., and shall be capable of passing the tests specified in 4.3.1 and 4.3.2. Glass shall be free from seeds, blisters, and other imperfections which may affect its appearance or impair its serviceability.

3.2 Design.

3.2.1 *Type I (size A.)* — Y-shaped tube shall be made of glass tubing having an outside diameter of 5.0 mm. ± ½ mm., with an average wall thickness of 1.0 mm. The Y-tube shall consist of three arms, each 60 mm. ± 2 mm. long, united in the form of a "Y". The ends shall be bulbous to insure the proper retention of ⅛ inch inside diameter rubber tubing, and shall be smoothly fire-polished without appreciable reduction in diameter.

3.2.2 *Type I (size B.)* — Y-shaped tube shall be made of glass tubing having an outside diameter of 8.0 mm. ± ½ mm., with an average wall thickness of 1.1 mm. The Y-tube shall consist of three arms, each 44 mm. ± 2 mm. long, united in the form of a "Y". The ends shall be bulbous to insure proper retention of ¼ inch inside diameter rubber tubing and shall be smoothly fire-polished without appreciable reduction in diameter.

3.2.3 *Type II.* — Right-angle connecting tube shall be made of glass tubing having an outside diameter of 8.0 mm. ± ½ mm., with an average wall thickness of 1.1 mm. The tubing shall be bent at a right angle, one arm approximately 63 mm. ± 2 mm. long, and other arm 57 mm. ± 2 mm. long. The bend shall not decrease the inside diameter of the tube more than ¾ mm. The ends shall be straight (untooled) but shall be smoothly fire-polished without appreciable reduction in diameter.

3.3 Workmanship.—Workmanship shall be first class throughout. Connecting tubes, both type I and type II, shall be smoothly and symmetrically finished. The glass shall be well annealed. When examined in a polariscope, there shall be no ring strain, and if any longitudinal strain is present, it shall be faint and highly diffused. Glass tubes shall be capable of withstanding frequent heat sterilizations without damage to any part, and shall require no special treatment after sterilization prior to use.

3.3.1 *Type I.*—Tubes shall be symmetrically made and properly joined.

3.3.2 *Type II.*—There shall be no flattening, bulging, discoloration or other imperfections which will affect the appearance or serviceability of the tube.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, samples shall be submitted in accordance with instructions furnished by the procuring activity (see 6.1). In addition, sampling procedures for the military agencies shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.2 Inspection.—Inspection may be made throughout the entire process of manufac-

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ture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery, as specified (see 6.1). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.2.1 Classification of defects (for military agencies only).

Categories and defects.

Major *AQL 2.50 percent defective.*

101. Not properly annealed.
102. Diameter of tubes not as specified.
103. Wall thickness not as specified.
104. Striations present.

Minor *AQL 6.5 percent defective.*

201. Length of tubes not as specified.
202. Tubes not symmetrical.
203. Tubes not properly joined.
204. Ends not fire-polished.
205. Seeds, blisters, chips and cracks present.

4.3 Tests.

4.3.1 *Resistance to shock.* — Glass tubes taken for test shall be placed in an oven which has previously been heated to 140° C., and allowed to remain therein at this temperature for at least 30 minutes. They shall then be removed from the oven and immediately plunged into water at a temperature of 4° C. Any breakage as a result of this test shall be cause for rejection.

4.3.2 *Solubility.* — Glass tubes taken for test shall meet the requirements for solubility described in paragraph entitled "Test for Containers of Type I" appearing in the current edition of the Pharmacopoeia of the United States.

5. PREPARATION FOR DELIVERY

The definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Unit of issue.*—One (1) tube, as specified, constitutes one unit.

5.1.2 *Quantities.*—Packaging and packing quantities shall be as specified in table I.

TABLE I.—*Packaging and packing quantities*

Federal stock number	Unit pack	Intermediate pack	Shipping container
6640–445–4085 6640–445–4100 or 6640–445–4105	1 unit	12 units	576 units

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in an envelope or carton of suitable size and design, or completely wrapped in kraft paper of suitable weight. Container shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.*—The number of units specified in table I shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Carton shall contain sufficient cushioning material to insure against shifting of, or damage to,

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the contents. Carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Tubes shall be protected and packaged in standard commercial containers of the size and kind commonly used for the purpose, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.* — The number of units specified in table I shall be overpacked in an exterior container designed for type 2 load and constructed in accordance with Federal Specifications PPP–B–585; style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping (when required) shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—The number of units specified in table I shall be overpacked in an exterior container conforming to Federal Specification PPP–B–601, table I, using class 1 plywood; or an exterior container designed for type 2 load and constructed in accordance with Federal Specifications PPP–B–621, table I; PPP–B–585, style 3, for class 2 use; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Civilian agencies.*—Marking shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*

5.3.2.1 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2.2 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Type and size required (see 1.2).

(*c*) Sampling required (see 4.1).

(*d*) Inspection required (see 4.2).

(*e*) Place of final inspection (see 4.2).

(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items appearing in the Armed Services Medical Stock List:

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6640–445–4085 TUBE, CONNECTING, LABORATORY GLASSWARE, Glass, Right Angle.

6640–445–4100 CONNECTOR, ELASTIC TUBING, BRANCHED, Glass, Y-Shaped, 5 mm. Outside Diameter.

6640–445–4105 CONNECTOR, ELASTIC TUBING, BRANCHED, Glass, Y-Shaped, 8 mm. Outside Diameter.

6.3 This specification includes the requirements of Military Specification MIL–T–15114 (BuMed), dated 15 February 1950.

6.4 **Transportation description.**—Transportation description applicable to this item is:

Glassware, chemical laboratory.

Carload minimum weight 10,000 pounds subject to Rule 34, Uniform Freight Classification.

Glassware, laboratory.

Motor volume minimum weight 18,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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